AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

MS102

Mid-term Examination MTA - Makeup
Fall Semester 2025/2026

Date: 6 /Dec. /2025

Number of Exam Pages:4 (including this cover sheet)	**Time Allowed:** 1.5hrs

Instructions:

- Total Marks: **30**
- This exam consists of **2** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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PART I: Answer the entire Multiple Choice Questions (MCQ). [15 Marks]

1. Which statement is true?
A. Distance can be less than displacement magnitude
B. Distance is always greater than or equal to the magnitude of displacement
C. Distance and displacement magnitude are always equal
D. Distance is always negative

2. How many significant figures are there in 0.004560?
A. 3
B. 4
C. 5
D. 6

3. The SI unit of force is:
A. Dyne
B. Newton
C. Joule
D. Pascal

4. The dimensional formula of velocity is:
A. $[L\ T^{-2}]$
B. $[L\ T^{-1}]$
C. $[M\ L\ T^{-1}]$
D. $[L^2\ T^{-2}]$

5. Which of the following quantities is dimensionless?
A. Velocity
B. Refractive index
C. Work
D. Density

6. When estimating, the process of rounding numbers to one significant figure is called:
A. Dimensional analysis
B. Approximation
C. Error minimization
D. Precision adjustment

7. The net force acting on an object at rest must be:
A. Zero
B. Equal to its mass
C. Maximum
D. Constant

8. The vector sum of two equal forces at right angles is:
A. Double either force
B. Half either force
C. $\sqrt{2}$ times either force
D. Zero

9. A car moves 100 m east and then 100 m west. Its displacement is:
A. 0 m
B. 100 m
C. 200 m
D. 50 m

10. The slope of a position–time graph represents:
A. Acceleration
B. Speed
C. Velocity
D. Displacement

11. The acceleration of an object is the rate of change of its:
A. Distance
B. Momentum
C. Velocity
D. Energy

12. Newton's First Law is also known as the Law of:
A. Momentum
B. Action and Reaction
C. Inertia
D. Acceleration

13. If the net force on a moving object is zero, the object will:
A. Stop immediately
B. Continue moving at constant velocity
C. Accelerate uniformly
D. Move in a circle

14. The acceleration of a freely falling object near Earth's surface is approximately:
A. 8.9 m/s²
B. 9.8 m/s²
C. 10.8 m/s²
D. 12.0 m/s²

15. When a ball hits a wall and bounces back, the momentum change is:
A. Zero
B. mv
C. 2mv
D. –mv

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16. Momentum is defined as:
A. Mass × Velocity
B. Force × Time
C. Work × Distance
D. Energy × Speed

17. The impulse-momentum theorem states that impulse equals:
A. Force × Velocity
B. Change in Momentum
C. Energy × Time
D. Work done

18. In a perfectly elastic collision, which of the following is conserved?
A. Only momentum
B. Only kinetic energy
C. Both momentum and kinetic energy
D. Neither

19. A 2 kg body moving at 3 m/s has momentum equal to:
A. 5 kg·m/s
B. 3 kg·m/s
C. 6 kg·m/s
D. 1.5 kg·m/s

20. If the acceleration of a body is zero, which of the following must be true?
A. A constant net force acts
B. The velocity is constant
C. The body is at rest
D. The mass is changing

21. The area under a velocity–time graph represents:
A. Acceleration
B. Displacement
C. Force
D. Work done

22. Two vectors of equal magnitude are opposite in direction. Their resultant is:
A. Equal to either
B. Zero
C. Double either
D. Half either

23. The SI unit of momentum is:
A. kg m/s
B. N m
C. N s
D. J/s

24. The equation $v^2 = u^2 + 2as$ is valid when:
A. Force is variable
B. Acceleration is constant
C. Acceleration is zero
D. Mass is changing

25. Which of these is a scalar quantity?
A. Velocity
B. Displacement
C. Speed
D. Force

26. A net force causes an object to:
A. Remain at rest
B. Change its motion
C. Lose mass
D. Gain potential energy only

27. The impulse delivered by a constant force equals the area under which graph?
A. Force–Distance
B. Force–Time
C. Velocity–Time
D. Acceleration–Time

28. During free fall, all objects (neglecting air resistance) have the same:
A. Mass
B. Acceleration
C. Velocity
D. Weight

29. The unit of work (joule) can also be expressed as:
A. N m
B. kg m/s
C. N s
D. m/s^2

30. In a collision where two bodies stick together, the collision is said to be:
A. Elastic
B. Inelastic
C. Partially elastic
D. Reversible

Part II: Answer the following questions. [15 Marks]

Question 1: (5 Marks)

Find the tension in the cord connecting two blocks m1 & m2 on horizontal surface. A force of 10.0 N is applied to the right on block m1. Assume a frictionless surface.

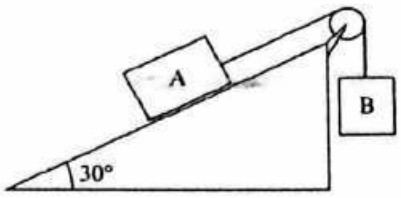
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The masses are m1 = 3.00 kg and m2 = 1.00 kg. Assume that the rope stays taut so that both blocks have the same acceleration.(20Marks)

Question 2: (5 marks)

Two packages are connected by a very light string that goes over an ideal pulley as shown in the figure. Package A has a mass of 3.0 kg and can slide along a rough plane inclined at 30° above the horizontal. The string acts on package A parallel to the surface of the plane. The coefficient of static friction between package A and the plane is 0.40. What minimum mass should package B have in order to start package A sliding up the ramp?



Question3 (5 marks)

A 4.00 kg object is moving at 5.00 m/s NO-RTHWEST. It strikes a 6.00 kg object that is moving SOUTHWEST at 2.00 m/s. The objects collide perfectly inelastically (stick together). The velocity of the 6.00 kg object after the collision is

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End of questions

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